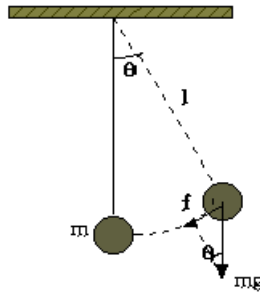


Introduction to Dimensional Analysis

Motivation



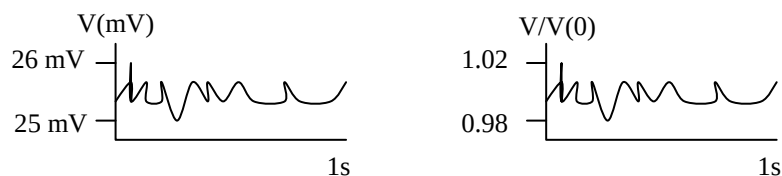
Q) In a simple pendulum, how does the period of oscillation τ depend on given physical parameters such as cord length l , gravitational acceleration g , and mass m ? If you double the length, does the oscillation double?

A) Physical intuition tells you that $\tau = f(l, g, m)$. Now, note that the dimension of τ is 1/(time). Similarly, the dimension of l , g , and m are (length), (length)/(time)², and (mass), respectively. Examination of the dimensions (units) leads to a conclusion that mass should be eliminated in the functional relation and that l and g should appear in the form of $\sqrt{l/g}$. Consequently, the relation should be $\tau \sim \sqrt{l/g}$ and doubling the cord length result in doubling the oscillation period.

- Dimension: General units to represent a physical quantity
M (mass), L (length), T (time), Θ (temperature)
Ex) velocity $\sim L/T$, force $\sim MLT^{-2}$
- Main idea: Relations between physical variables should be dimensionally homogeneous. In other words, they should be invariant under the change of units (principle of dimensional invariance). Otherwise, the equation is unit-dependent. The dimensional invariance implies that any physical relation can be expressed in terms of dimensionless variables.
 - For example, the law for gravitational force $F = mg$ is dimensionally homogeneous (can use any consistent units). However, $F = 9.8m$ is dimension-dependent (m should be in kg and F in N).

- $u = \left(\frac{dp}{dx} \right) / \mu$ should not be a correct equation because it is not dimensionally homogeneous.
 - A relation $f(a, b, c, \dots) = 0$ can be transformed into $g(\alpha, \beta, \dots) = 0$, where $a, b, c, \dots$ are dimensional variables while $\alpha, \beta, \dots$ are dimensionless. The number of dimensionless variables should be smaller than that of dimensional variables in this transformation.
- Dimensional analysis: Dimensional analysis is to express a relation in terms of dimensionless variables, instead of using dimensional variables. The relation may be just a statement describing functional dependence of a certain variable on others. Also, the relation could be represented by a governing equation.
 - For example, compare the following two statements: (1) The drag force applied on a sphere exposed to fluid flow is a function of flow velocity, viscosity, density, and diameter of the sphere, i.e., $F_D = f(V, \mu, \rho, D)$. (2) The dimensionless drag force is a function of some dimensionless variable named Reynolds number, i.e., $\frac{F_D}{\rho D^2 V^2} = f(\text{Re})$, where Reynolds number $\text{Re} \equiv \frac{\rho V D}{\mu}$. Note that dimensional analysis has reduced the number of variables from 5 (case1) to 2 (case2) in this example.
 - As another example, compare the momentum equations for steady two-dimensional flow in two different forms: $\rho \left(u \frac{\partial u}{\partial x} + v \frac{\partial u}{\partial y} \right) = \mu \left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} \right)$, $u^* \frac{\partial u^*}{\partial x^*} + v^* \frac{\partial u^*}{\partial y^*} = \frac{1}{\text{Re}} \left(\frac{\partial^2 u^*}{\partial x^{*2}} + \frac{\partial^2 u^*}{\partial y^{*2}} \right)$, where $u^* \equiv \frac{u}{V}$, $v^* \equiv \frac{v}{V}$, $x^* \equiv \frac{x}{L}$, $y^* \equiv \frac{y}{L}$. In this case, non-dimensionalization leads to an idea that Re is an important parameter that determines the dynamics of the fluid motion. It will be later shown that Re is a very important parameter that gives the relative magnitude of inertia force compared with the viscous force.

- In general, dimensionless expressions have the following advantages:
 1. Non-dimensionalization reduces the number of associated variables describing a physical relation. This is because the principle of dimensional homogeneity permits only certain combinations of the dimensional variables to form dimensionless variables. In experimental fluid mechanics and heat transfer, this reduction is very useful. Let's consider the example of drag force measurement. As shown in the expression (1), 10^4 experiments have to be performed to determine the drag force for combinations of 10 different velocities, viscosities, densities, and diameters. On the other hand, if we use dimensionless variables, the dimensionless drag force can be measured just for 10 different Reynolds numbers.
 2. Nondimensionalization of a governing equation often simplifies the equation and provides useful information (important parameters, e.g., Re , relative importance of each terms).
 3. It is often easier to interpret the dimensionless representation of a phenomenon (easier to know the physical meaning) because physical quantities associated with the original problem are used as units to describe the phenomenon. For example, let's consider a graph displaying a noisy electrical signal. It is easier to look at the dimensionless plot on the right-hand side (normalized by its initial voltage) to determine how noisy the signal is. Our previous study on transient heat conduction tells us that for transient heat conduction problems, especially for lumped heat capacitance problems, L_c^2/α (characteristic length squared divided by thermal diffusivity) is a good (physically given) unit for time.



4. Dimensionless equations good for scaling up or scaling down of the results.
5. In numerical computation, use of dimensionless variables removes possible round-off errors caused by dealing with two large/small numbers.